CUNEF UNIVERSIDAD

Econometrics 2024/2025

Topic 0: Review of Statisics

Outline

Mean, median and mode

Variance and standard deviation

Covariance and correlation

Regression line

Probability distributions: Normal and Student's t

Point estimation

Hypothesis tests

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Arithmetic Mean

- The arithmetic mean (or simply mean) of a set of data is the sum of the data values divided by the number of observations.
- Population mean (if we have the entire population available):

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N} = \frac{x_1 + x_2 + \ldots + x_N}{N}$$

  where μ is the parameter.
- If the data set is from a sample, then the sample mean $\bar{x}$ is a statistic:

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N} = \frac{x_1 + x_2 + \ldots + x_N}{N}$$

- If the same observation repeats several times:

$$\sum_{i=1}^{N} x_i = \sum_{i=1}^{N} n_i \cdot x_i$$

Arithmetic Mean

- Arithmetic mean is the most intuitive and commonly used measure for central tendency.
- It can be applied for **quantitative data only**. It is impossible to calculate arithmetic mean for qualitative data.
- It has a major disadvantage: it is not robust to outliers.

Robust: strong, not-sensitive to small changes or variations.
Outlier: atypical data point.

Properties of the arithmetic mean

- Call $Y = a + b \cdot X$ a function of X, where a and b are some numbers.
- Then we say that Y is a linear function of X.
- If we know the mean of X, then the mean of Y is just $\bar{y} = a + b \cdot \bar{x}$.

Weighted mean

- Some situations require a special type of mean called a weighted mean.
- For example, in calculating the final grade for this course.
- Or when we merge several groups together and want to calculate the overall mean.

$$W = \frac{\sum n_i x_i}{N} = \sum w_i x_i,$$

where $w_i = n_i/N$ and $\sum n_i = N$.

Median

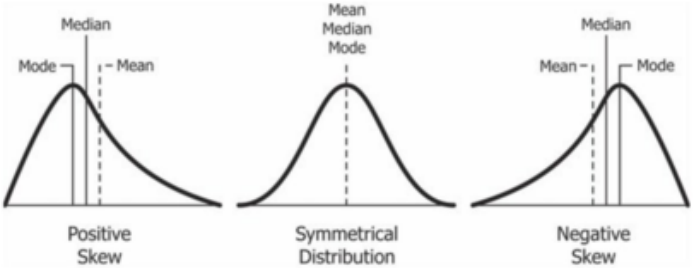
- The median is the middle observation of a set of observations that are **ordered** in increasing order.
- If the sample size, n, is an even number, the median is the average of the two middle observations.
- The median can be found in $0.5(n+1)th$ ordered position. If the position is with a decimal (like .5), take the mean of the two middle observations.
- In many cases median is preferred because it is robust to outliers.

Mode

- The mode, if one exists, is the most frequently occurring value.
- A distribution with one mode is called unimodal; with two modes, it is called bimodal; and with more than two modes, the distribution is said to be multimodal.
- The mode is most commonly used with qualitative data.

Mean, median, mode: the shape of the distribution

- Comparison of mean, median and mode also help to describe the shape of the distribution.



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Variance and Standard Deviation

- The variance is the average of the sum of squared distances from the mean.
- The population variance σ^2 is the sum of the squared differences between each observation and the *population* mean divided by the *population* size N:

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N} = \frac{\sum_{i=1}^{N} x_i^2}{N} - \mu^2$$

- The sample variance [1] s^2 is the sum of the squared differences between each observation and the *sample* mean divided by the *sample* size N minus one:

$$s^2 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1} = \frac{N}{N-1}\sigma^2$$

- To compute the variance requires squaring the distances, which then changes the unit of measurement to square units. The standard deviation, which is the square root of variance, restores the data to their original measurement unit.

$$\sigma = \sqrt{\sigma^2} \qquad s = \sqrt{s^2}$$

[1] ometimes called a quasi-variance

Variance and Standard Deviation: properties

- σ^2 and σ are always ≥ 0
- If $Y = a + b \cdot X$ and we know σ_x^2, then:

$$\sigma_y^2 = b^2 \cdot \sigma_x^2$$
$$\sigma_y = \mid b \mid \cdot \sigma_x$$

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Covariance

- Covariance is numerical a measure of the **linear** relationship between two variables.
- A population covariance is given by

$$Cov(x, y) \equiv \sigma_{xy} = \frac{\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)}{N} = \frac{\sum x_i y_i}{N} - \mu_x \mu_y$$

- A sample covariance is given by

$$Cov(x, y) \equiv s_{xy} = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{N-1} = \frac{N}{N-1}\sigma_{xy}$$

- If $s_{xy} > 0$, then X and Y have a positive linear relationship; if $s_{xy} < 0$, then X and Y have a negative linear relationship; if $s_{xy} = 0$, then X and Y do no have a linear relationship (might still have a non-linear relationship).
- **Covariance does not provide a measure of the strength of the relationship**.

Correlation

- A correlation coefficient provides the direction (positive, negative) and strength (weak, moderate, strong).
- A population correlation coefficient:

$$\rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

- A sample correlation coefficient:

$$r = \frac{s_{xy}}{s_x s_y}$$

Correlation

- The correlation coefficient ranges from -1 to $+1$.
- r close to $+1$ indicates a positive linear relationship.
- r close to -1 indicates a negative linear relationship.
- When $r = 0$, there is no linear relationship.

If the two random variables are independent, the covariance/correlation will be zero. However, a covariance/correlation of zero does not necessarily mean that the variables are independent. A nonlinear relationship might exist.

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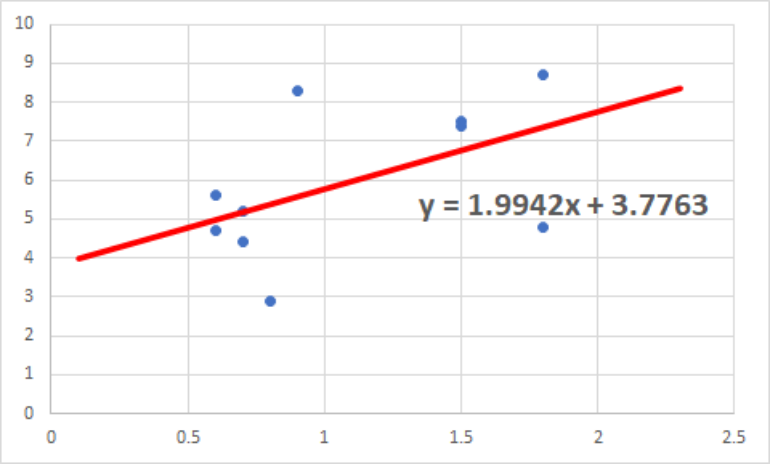
- Finally, we might want to **explain** the relationship between the two variables in economic terms.
- Also, we might want to **predict** the value of one variable given the value of another variable.
- For example, if I know the number of views in CANVAS I can predict the student's grade.
- For this we can draw a line through the scatterplot that **fits the data best**.
- Such line is called a regression line.

Regression line



Regression line

- In order to calculate the regression line, we have to calculate the slope and the intercept:

$$Y = a + b \cdot X,$$

  where Y is the dependent variable and X is the explanatory variable.
- The slope and the intercept are given by:

$$b = \frac{\sigma_{xy}}{\sigma_x^2}, \qquad a = \bar{y} - b \cdot \bar{x}$$

- The goodness of fit is measured by the coefficient of determination:

$$R^2 = r^2$$

Regression line

- The coefficients of the regression line can be interpreted as follows:
  - When X increases (decreases) by one unit, Y will increase (decrease) by b units.
  - When the value of X is equal to zero, Y will be equal to a.
- Given the regression line, we can also make predictions by plugging in a value of X into the equation.
- R^2 varies from 0 to 1. R^2 measures the proportion of variation in the dependent variable (Y) that can be attributed to the independent variable (X).
  - If R^2 is close to 0, then the goodness of fit is bad and the explanatory power of X is low.
  - If R^2 is close to 1, then the goodness of fit is good and the explanatory power of X is high.

Regression line: example

Consider investigating how years of education affect income.

- Which variable is the dependent variable Y?
- Which variable is the explanatory variable X?
- What relationship do you expect to find?

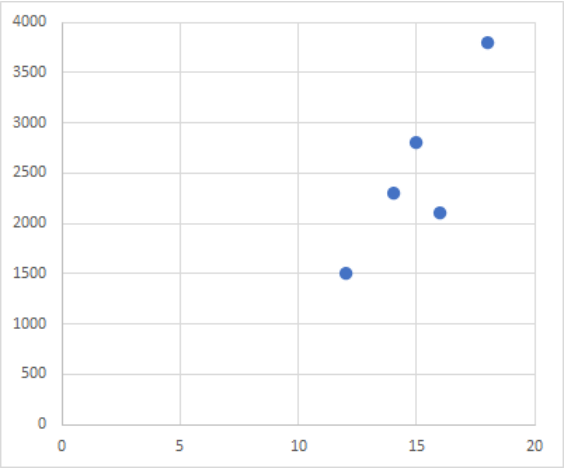
Regression line: example

Now you collect some data about the years of education and monthly income of five people.

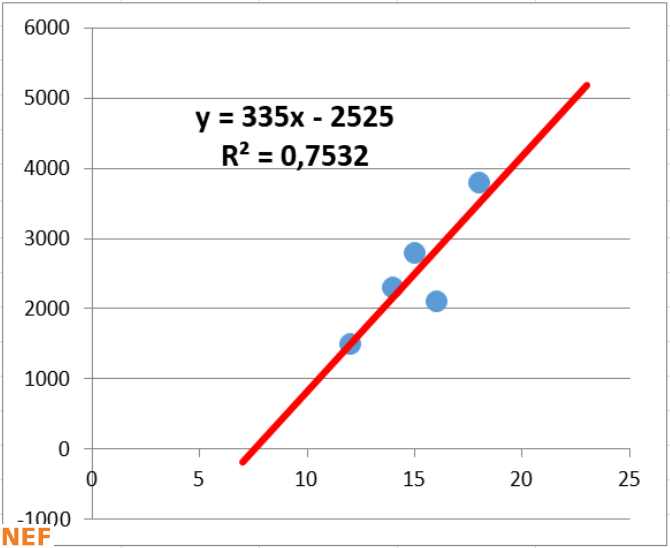
	years of education (X)	monthly income ($) (Y)	x*y	x^2	y^2
	12	1500	18000	144	2250000
	14	2300	32200	196	5290000
	16	2100	33600	256	4410000
	15	2800	42000	225	7840000
	18	3800	68400	324	14440000
SUM	**75**	**12500**	**194200**	**1145**	**34230000**
n=	**5**				

- How does an increase in one year in education affect income?
- What income would you predict for a person who has 17 years of education (12 years of high school + 5 years of double university degree)?
- Evaluate the goodness of fit.

Regression line: example



Regression line: example



mean x	15	years
mean y	2500	$
var x	4	years^2
var y	596000	$^2
cov	1340	years*$
cor	0,87	-
b	335	
a	-2525	
R2	0,75	75%

Regression line: example

$$Y = -2525 + 335 \cdot X$$

- How does an increase in one year in education affect income?
  For an increase in one year of education (X), income will increase by 335\$.
- What income would you predict for a person who has 17 years of education (12 years of high school + 5 years of double university degree)?
  For a person who has had 17 years of education the predicted average income is 3170\$.
- Evaluate the goodness of fit. R^2 has the value of 75%, which indicates a good fit. In other words, the variable years of education (X) can explain 75% of variation in the variable income (Y).

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Normal distribution

- By far the most popular and widely used of all continuous distributions is the Normal (Gaussian, bell-shaped) distribution.
- It can take **any** value on the real line, i.e. $-\infty, +\infty$.
- Variables that are approximately Normally distributed: height, weight, shoe size, IQ, stock returns, average grades, log(income), etc.
- In nature: snowflake sizes, temperatures, birth weights of any animal, etc.
- Normal distribution is also the keystone of econometric and statistic theory, as it is the basis of the Central Limit Theorem.

Normal distribution

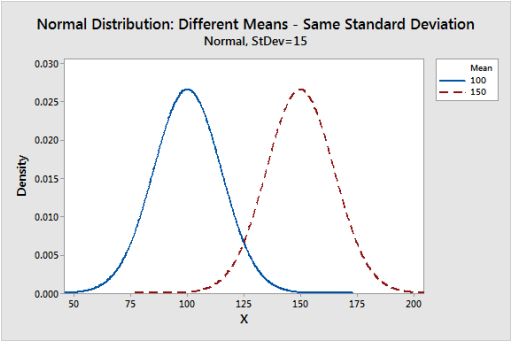
- If a r.v. X is Normally distributed, i.e. $X \sim N(\mu, \sigma^2)$, then its PDF is given by:

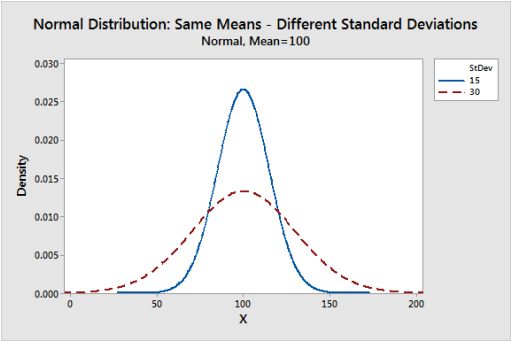
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

- The mean and variance are given by:

$$E[X] = \mu, \qquad V[X] = \sigma^2$$

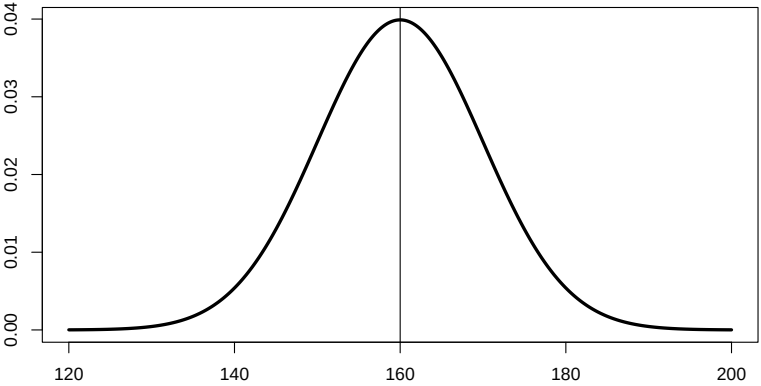
Normal distribution





Normal distribution: example

Call X - students' height (in cm) at CUNEF. We assume it is Normally distributed $X \sim N(\mu = 160, \sigma^2 = 100)$. What is the probability that a randomly selected student is shorter than 170cm?



Normal distribution: example

If $X \sim N(\mu = 160, \sigma^2 = 100)$, then $P(X < 170)$:

- Manually:

$$P(X < 170) = F(170) = \int_{-\infty}^{170} \frac{1}{\sqrt{2\pi \cdot 100}} \exp\left\{\frac{(x-160)^2}{2 \cdot 100}\right\} dx = ?$$

- Use `R`: `pnorm(170,160,sqrt(100))`
- Use statistical tables.

Normal distribution: statistical tables

- Statistical tables facilitate the calculation of the complicated integrals:

$$P(X < x) = F(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{(x-\mu)^2}{2\sigma^2}\right\} dx$$

- But... the statistical tables are calculated for standard Normally distributed variables only: $X \sim N(0,1)$.

Normal distribution: statistical tables

X standard Normal $X \sim N(0,1)$. What is the $P(X < 0.75)$?

$$P(X < 0.75) = F(0.75) = \int_{-\infty}^{0.75} \frac{1}{\sqrt{2\pi \cdot 1}} \exp\left\{\frac{(x-0)^2}{2 \cdot 1}\right\} dx = 0.7734$$

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	[illegible]	0.8051	0.

In R: pnorm(0.75,0,1) $= 0.7733726$

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Normal distribution: standardization

- And if the variable is not standard Normally distributed, we need to standardize it first! And only then we can use the table.

$$Z = \frac{X - \mu}{\sigma}$$

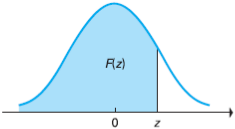
- For example, X is students' height (in cm) at CUNEF. We assume it is Normally distributed $X \sim N(\mu = 160, \sigma^2 = 100)$. What is the probability that a randomly selected student is shorter than 170cm?

$$P(X < 170) = P\left(\frac{X - \mu}{\sigma} < \frac{170 - \mu}{\sigma}\right) = P\left(Z < \frac{170 - 160}{10}\right) = P(Z < 1)$$

Normal distribution: standardization

Then $P(X < 170) = P\left(\frac{X-\mu}{\sigma} < \frac{170-160}{10}\right) = P(Z < 1) = 0.8413$

Table 1 Cumulative Distribution Function, $F(z)$, of the Standard Normal Distribution Table



z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	[illegible]	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015

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Student's t distribution

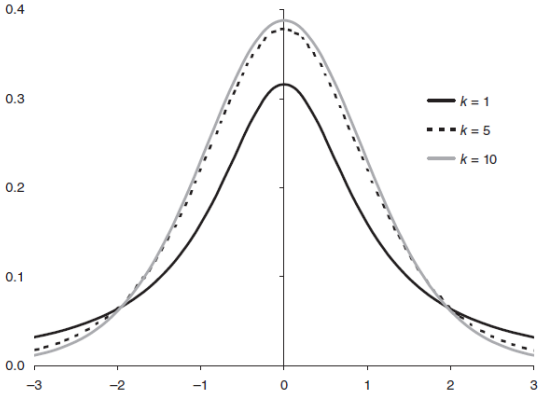
- The Student's t distribution is extremely popular distribution in statistics and in risk management.
- was first described in English, in 1908, by William Sealy Gosset, an employee at the Guinness brewery in Dublin.
- The probability density function can be written:

$$f(x) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi}\,\Gamma\left(\frac{k}{2}\right)}\left(1+\frac{x^2}{k}\right)^{-\frac{(k+1)}{2}},$$

where k is the degrees of freedom and Γ is the gamma function.

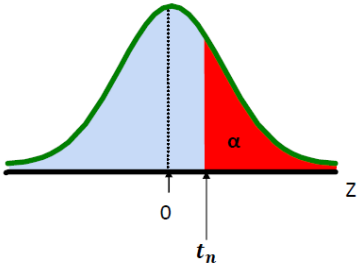
Student's t distribution

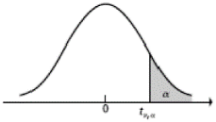
- The t distribution is symmetrical around its mean, which is equal to zero. For low values of k, the t distribution looks very similar to a standard normal distribution, except that it displays excess kurtosis. As k increases, this excess kurtosis decreases. As k approaches infinity, the t distribution converges to a standard normal distribution.



Student's t distribution

Example: Calculate the following probability: $P(t_4 > 2{,}132) = ?$





La tabla muestra, para algunas probabilidades α, los valores de la $t^{\nu}_{\alpha,n}$ tales que $P(t_\nu > t_{\nu,\alpha}) = \alpha$, donde t_ν es una variable aleatoria t de Student con ν grados de libertad. Por ejemplo, la probabilidad de que una variable aleatoria t de Student con 10 grados de libertad sea mayor que 1,372 es 0,10.

ν	α				
	0,100	0,050	0,025	0,010	0,005
1	3,078	6,314	12,706	31,821	63,657
2	1,886	2,920	4,303	6,965	9,925
3	1,638	2,353	3,182	4,541	5,841
4	1,533	2,132	2,776	3,747	4,604
5	1,476	2,015	2,571	3,365	4,032
6	1,440	1,943	2,447	3,143	3,707
7	1,415	1,895	2,365	2,998	3,499
8	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

$P(t_4 > 2.132) = 0.05$

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Point estimation

- Suppose we want to estimate the average income μ of all families in a city.
- It seems logical to use the sample mean $\bar{X}$ as an estimator of the population mean μ.
- We need to select a random sample (for example $n = 80$), and then we would obtain the average income of the sample, for example $\bar{x} = 800$.
- Then, the estimator of the population mean μ will be $\widehat{\mu} = \bar{X}$, that is, the mean statistic will be $\overline{\mathrm{X}}$ and the point estimate will be $\widehat{\boldsymbol{\mu}} = \overline{\boldsymbol{x}} = 800$.
- The problem is that there are many different ways to estimate the average income μ. Using the sample mean $\bar{X}$ is only one of them.
- This makes it necessary to establish standards or criteria that allow choosing which is the "best" estimator among the possible ones in each case.

Mean square error of the estimator

- As the parameter is unknown, we will never know exactly to what extent each estimate (approximation) is far or close to the value of the parameter, that is, its use leads to the possibility of making a more or less high error when working with the point estimate as if it were the true value.
- We define the mean square error of the estimator $\hat{\theta}$, we denote as $\mathrm{MSE}(\hat{\theta})$, as the expected value of the square of the difference between the estimator $\hat{\theta}$ and the parameter θ, that is

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{E}\left[(\hat{\theta} - \theta)^2\right]$$

- The smaller the error, the more accurate the estimate

Mean square error of the estimator

- An alternative expression of the mean squared error is:

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \text{VAR}(\hat{\theta}) + (\underbrace{\mathbb{E}[\hat{\theta}] - \theta}_{\text{bias}[\hat{\theta}]})^2$$

- The MSE is equal to the variance of the estimator $\text{VAR}(\hat{\theta})$, plus the square of the bias of the estimator: $E[\hat{\theta}] - \theta = \text{bias}[\hat{\theta}]$.
- Ideally, both quantities must be as small as possible, which is equivalent to the fact that the sampling distribution of $\hat{\theta}$ is concentrated around the value of the parameter θ. The more concentrated, the lower the variance.

Bias, efficiency, and consistency of an estimator

- **Unbiased estimator**: $\hat{\theta}$ is an unbiased estimator of the parameter θ if the expectation of the estimator $\hat{\theta}$ is equal to the parameter θ, that is:

$$E[\hat{\theta}] = \theta$$

  Alternatively,

$$\text{bias}(\hat{\theta}) = \mathrm{E}[\hat{\theta}] - \theta = 0$$

  Otherwise we will say that the estimator is biased when $\text{bias}(\hat{\theta}) \neq 0$.
- **Efficient estimator**: The most efficient estimator among a group of estimators will be the one with the smallest variance.
- **Consistent estimator**: An estimator of a population θ is consistent when its variance tends to zero as n increases.

$$\lim_{n \to \infty} \text{Var}(\widehat{\theta}) = 0$$

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Null and Alternative Hypotheses

- In this topic, we will review how to perform statistical tests to evaluate different hypotheses about a population.
- Our starting point will be the null hypothesis:

$$\mathrm{H}_0 : \mu = \mu_0.$$

- In contrast the alternative hypothesis, H_1, will take one of three forms, i.e. using $\neq, <$ or $>$, that is:

$$\mathrm{H}_1 : \mu \neq \mu_0 \quad \text{or} \quad \mathrm{H}_1 : \mu < \mu_0 \quad \text{or} \quad \mathrm{H}_1 : \mu > \mu_0$$

  Note that only one of these forms will be used per test. To determine which form to use will require careful consideration of the wording in the problem.

Null and Alternative Hypotheses

- How to correctly interpret a hypothesis test?
- Remember that the problem is to use evidence in a randomly selected sample of data to decide whether to accept the null H_0 or to reject it in favor of the alternative H_1.
- If the null is accepted, this does not mean that the null can de declared "true".
- Rather, it is accepted tentatively with the recognition that it might be rejected later based on additional evidence.

Tests for normal populations

- Let us suppose that, having obtained sample data from a normal distribution, we use the estimator $\widehat{\theta}$ to estimate an unknown parameter θ.
- Assume $\widehat{\theta}$ has the sampling distribution:

$$\widehat{\theta} \sim N(\theta, V)$$

  where $V = \text{Var}(\widehat{\theta})$ is assumed to be known.
- Hence the standard error, S.E. $(\widehat{\theta})$, is also known.

Tests for normal populations

The hypothesis test would proceed using the following steps:

- 1. Define the hypotheses. In this case suppose we test:

$$\mathrm{H}_0 : \theta = \theta_0 \quad \text{vs.} \quad \mathrm{H}_1 : \theta > \theta_0.$$

- 2. State the test statistic and its distribution, then compute its value. We know that standardization of $\widehat{\theta}$ yields:

$$\frac{\widehat{\theta} - \theta_0}{\text{S.E.}\ (\widehat{\theta})} \sim N(0, 1).$$

Next compute the test statistic value using the sample data, i.e. $\widehat{\theta}$ will be our point estimate of θ.

Tests for normal populations

- 3. Define the critical region for a given significance level, α. For example, let $\alpha = 0.05$, where α is the Type I error probability. Since we are conducting an upper-tailed test, we require $z_\alpha = z_{0.05} = 1.645$, i.e. the z-value above which lies 5% probability.

- 4. Decide whether or not to reject H_0. If the test statistic value lies in the critical region then we will reject H_0, otherwise we do not reject H_0.

Remember that we will be always using a test statistic of the form:

$$\frac{\text{point estimator - hypothesised value}}{\text{(estimated) standard error}}.$$

Although this is not the format of all test statistics, it does cover a wide class of situations.

p-values

- So, given we reject H_0 if the test statistic value is sufficiently extreme, we would also reject H_0 if we obtained a sufficiently small *p*-value.
- To summarize, in the *p*-value world of testing, we use the following simple decision rule.

p-value decision rule

- Reject H_0 if p-value $< \alpha$, where α is the significance level.
- Do not reject H_0 if p-value $\geq \alpha$.

Calculating p-values

Let the test statistic value be z, and the test statistic (a random variable as it is constructed from an estimator) be Z.

Assuming a test statistic distribution which is symmetric about zero for the two-tailed test, in general these can be summarized as follows.

Form of H_1	p-value calculation
$\mathrm{H}_1 : \theta \neq \theta_0$	$2 \times P(Z \geq \lvert z \rvert)$
$\mathrm{H}_1 : \theta < \theta_0$	$P(Z \leq z)$
$\mathrm{H}_1 : \theta > \theta_0$	$P(Z \geq z)$

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